

8) Prove the idempotent laws in Table 1 by showing that

a) $A \vee A = A$ b) $A \wedge A = A$

Solution: a) Let, x be an arbitrary element of $A \vee A$, i.e. $x \in A \vee A$.

By Defn. 1, $x \in A \vee x \in A$. By defn. of OR, ~~at least~~ at least one of $x \in A$ and $x \in A$ is True. WLOG, Let $x \in A$ be True.

By Defn. 3 of Section 2.1, $A \vee A \subseteq A$.

Let x be an arbitrary element of A , i.e. $x \in A$.

By Defn. of OR, $x \in A \vee x \in A$ is True. By Defn. 1, $x \in A \vee A$.

By Defn. 3 of Section 2.1, $A \subseteq A \vee A$.

$\therefore A \vee A = A$ $\square$

b) Let, x be an arbitrary element of $A \wedge A$, i.e. $x \in A \wedge A$.

By Defn 2, $x \in A \wedge x \in A$. By defn. of AND, both $x \in A$ and $x \in A$ are True.

By Defn. 3 of Section 2.1, $A \wedge A \subseteq A$.

Let, x be an arbitrary element of A , i.e. $x \in A$.

By Defn. of AND, $x \in A \wedge x \in A$ is True. By Defn. 2, $x \in A \wedge A$.

By Defn. 3 of Section 2.1, $A \subseteq A \wedge A$.

$\therefore A \wedge A = A$ $\square$

Prove the complement laws in Table 1 by showing that a) $A \vee \bar{A} = 1$